MILITARY SPECIFICATION

CARTRIDGE, IGNITION, M702A1
PARTS FOR

This Specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions for the fabrication of parts and packing of parts for M702A1, Ignition Cartridge for the 60MM Lightweight Company Mortar System.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC: 1310

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

9347046 - Body
12630942 - Plug, Firing
12630943 - Tube
12630945 - Cap
12630946 - Tube, Flash
12630947 - Head

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on Drawings (dwgs) 9347046, 12630942, 12630943, 12630945, 12630946, 12630947 as applicable, and with all requirements specified in applicable specifications and standards.

3.3 Bursting strength. The tube (12630943) shall not burst at a compression load of 57 pounds, minimum, but shall burst at a compression load of 192 pounds, maximum. Testing shall be as specified in 4.5.2.

3.4 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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3.5 Workmanship. All parts shall be fabricated and finished in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the components in the quantities described in Table I.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48653(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body (dwg. 9347046) Examination for defects	25		3.2	4.4.2.1
	Plug, Firing (dwg. 12630942) Examination for defects	25		3.2	4.4.2.2
	Tube (dwg. 12630943) Examination for defects Bursting strength test	25 25		3.2 3.3	4.4.2.3 4.4.2.3/4.5.2
	Cap (dwg. 12630945) Examination for defects	25		3.2	4.4.2.4
	Tube, Flash (dwg. 12630946) Examination for defects	25		3.2	4.4.2.5
	Head (dwg. 12630947) Examination for defects	25		3.2	4.4.2.6

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.1	Body				9347046
					NEXT HIGHER ASSEMBLY
					12630941
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special A	Flash hole missing or obstructed		100%	3.2	Visual
Major					
101	Pitch diameter of threads, min.		0.40%	3.2	Gage
102	Major diameter of threads, min.		0.40%	3.2	Gage
103	Diameter of primer cavity		0.40%	3.2	Gage
104	Depth of Primer cavity		0.40%	3.2	Gage
105	Depth of flash tube cavity		0.40%	3.2	Gage
106	Diameter of flash tube cavity		0.40%	3.2	Gage
107	Diameter of flash hole		0.40%	3.2	Gage
108	Protective finish missing		0.40%	3.2	Visual
109	Large outside diameter		0.40%	3.2	Gage
110	Diameter of pellet cavity		0.40%	3.2	Gage
111	Depth of pellet cavity		0.40%	3.2	Gage
112	True position of primer cavity		0.40%	3.2	Gage
113	Length of sealant groove		0.40%	3.2	Gage
114	Depth of sealant groove		0.40%	3.2	Gage
115	Small outside diameter		0.40%	3.2	Gage
116	Diameter of countersink at base of primer cavity, max.		0.40%	3.2	Gage
117	Length from large diameter to small end		0.40%	3.2	Gage
118	True position of outside diameter		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48653(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Body	2 OF 2	9347046
			NEXT HIGHER ASSEMBLY: 12630941

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
119	Intermediate outside diameter		0.40%	3.2	Gage
120	Length from large outside diameter to end of external thread		0.40%	3.2	Gage
121	Length from large diameter to beginning of thread		0.40%	3.2	Gage
122	Length of large diameter section		0.40%	3.2	Gage
Minor					
201	Pitch diameter of thread, max.		0.65%	3.2	Gage
202	Major diameter of thread, max.		0.65%	3.2	Gage
203	Radii, chamfer and countersink missing or incorrect		0.65%	3.2	Gage
204	Surface finish improper		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48653(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Plug, Firing	1 OF 1	12630942
			NEXT HIGHER ASSEMBLY: 12630941

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length of tip		0.40%	3.2	Gage
102	Total length including tip		0.40%	3.2	Gage
103	Length from tip to small diameter		0.40%	3.2	Gage
104	True position of tip		0.40%	3.2	Gage
105	Radius improper		0.40%	3.2	Gage
106	Large diameter		0.40%	3.2	Gage
107	Small diameter		0.40%	3.2	Gage
Minor					
201	Protective finish improper		0.65%	3.2	Visual
202	Chamfer missing		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.3	Tube		1 OF 1	12630943
				NEXT HIGHER ASSEMBLY
				9347044

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bursting strength test		(a)	3.3	4.5.2
102	Thickness		0.40%	3.2	Gage
103	Length		0.40%	3.2	Gage
104	Inside diameter		0.40%	3.2	Gage
105	Tube not properly spiral wound		0.40%	3.2	Visual
106	Evidence of tube delaminations, holes, or separations		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:
(a) Sample per MIL-STD-105, Inspection Level I, 0.40% AQL.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Cap	1 OF 1	12630945
			NEXT HIGHER ASSEMBLY: 12630941

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	True position of cavity		0.40%	3.2	Gage
103	Diameter of cavity		0.40%	3.2	Gage
104	Depth of cavity		0.40%	3.2	Gage
105	Length from base of cavity to face		0.40%	3.2	Gage
106	Protective finish missing		0.40%	3.2	Visual
107	Height of flange (3 places)		0.40%	3.2	Gage
Minor					
201	Radii or chamfers missing		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48653(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630946
4.4.2.5	Tube, Flash				NEXT HIGHER ASSEMBLY 12630944
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
A	Flash hole(s) missing or obstructed		100%	3.2	Visual
B	Longitudinal flash hole missing or obstructed		100%	3.2	Visual
Major					
101	True position of flash holes		0.40%	3.2	Gage
102	Diameter of flash holes		0.40%	3.2	Gage
103	Length		0.40%	3.2	Gage
104	Large outside diameter		0.40%	3.2	Gage
105	Small outside diameter		0.40%	3.2	Gage
106	Length of small outside diameter		0.40%	3.2	Gage
107	Inside diameter		0.40%	3.2	Gage
108	Length of inside diameter (pin end)		0.40%	3.2	Gage
109	Inside diameter (pin end)		0.40%	3.2	Gage
110	Protective finish missing or damaged		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48653(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.6	Head		12630947
			NEXT HIGHER ASSEMBLY 12630941

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of internal threads, max.		0.40%	3.2	Gage
102	Minor diameter of internal threads, max.		0.40%	3.2	Gage
103	Pitch diameter of external threads, min.		0.40%	3.2	Gage
104	Major diameter of external threads, min.		0.40%	3.2	Gage
105	Length of internal thread, min.		0.40%	3.2	Gage
106	Overall length		0.40%	3.2	Gage
107	Length of large internal diameter		0.40%	3.2	Gage
108	Small internal diameter		0.40%	3.2	Gage
109	Large internal diameter		0.40%	3.2	Gage
110	Large outside diameter		0.40%	3.2	Gage
111	Small outside diameter		0.40%	3.2	Gage
112	Protective finish missing		0.40%	3.2	Visual
113	True position of internal thread		0.40%	3.2	Gage
114	True position of small internal diameter		0.40%	3.2	Gage
115	True position of large internal diameter		0.40%	3.2	Gage
Minor					
201	Pitch diameter of internal thread, min.		0.65%	3.2	Gage
202	Pitch diameter of external thread, max.		0.65%	3.2	Gage
203	Major diameter of external thread, max.		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48653(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 12630947
4.4.2.6	Head				NEXT HIGHER ASSEMBLY 12630941
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
204	Minor diameter of internal thread, min.		0.65%	3.2	Gage
205	Length from small end to shoulder		0.65%	3.2	Gage
206	Length of external thread undercut		0.65%	3.2	Gage
207	Length from shoulder to end of external thread		0.65%	3.2	Gage
208	Radii, chamfers or countersinks missing or incorrect		0.65%	3.2	Visual
209	Surface finish improper		0.65%	3.2	Visual
210	Depth of spanner holes		0.65%	3.2	Gage
211	True position of spanner holes		0.65%	3.2	Gage
212	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manfacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Bursting strength. The tube shall be cut in half lengthwise and one of the halves placed in the fixture (see figure 1). The stop (3) shall be completely removed or backed off so that it does not touch the punch (2). Place fixture in universal tester set for a travel speed of 0.125 inches per minute. Proceed with Test Number 1 and 2.

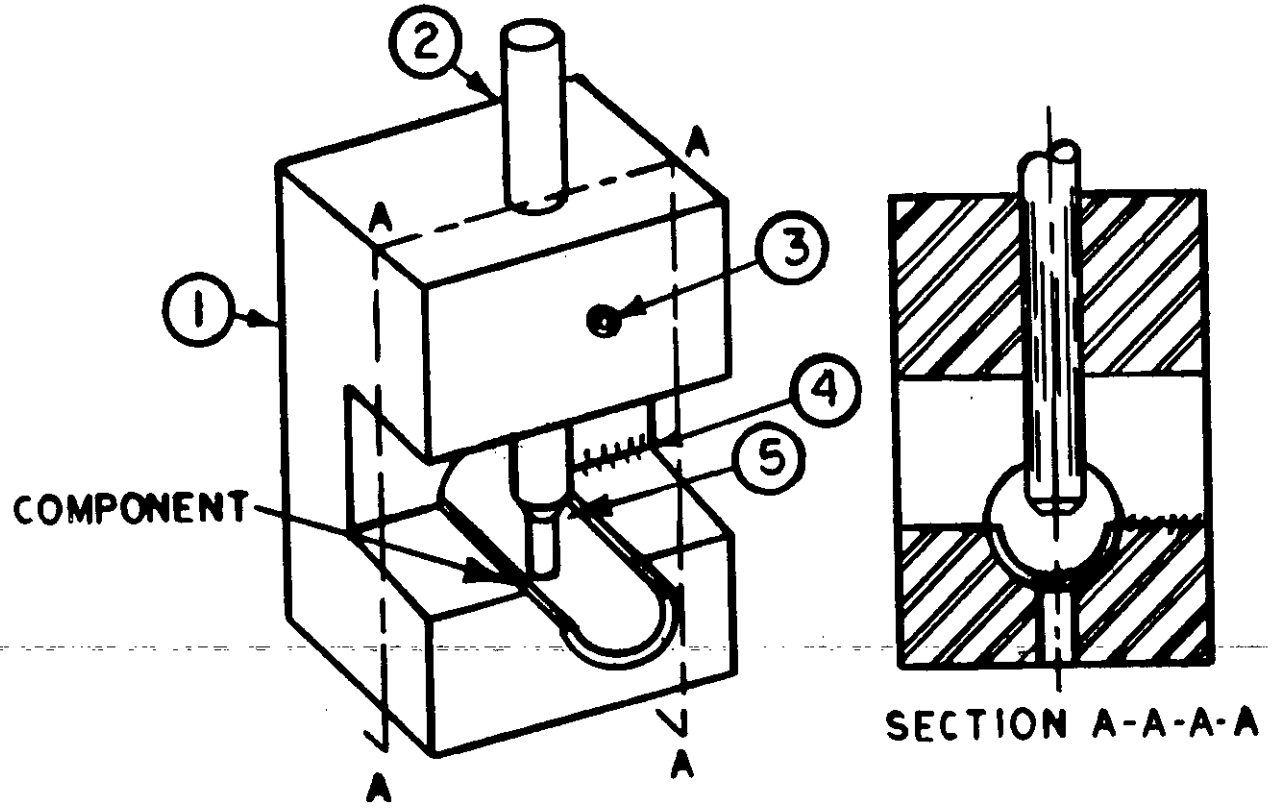
TEST NUMBER 1: NO BURST

Apply 57 pound compression load, remove body assembly, Item is rejected if evidence of bursting is observed. Tearing or puncturing of material shall be considered evidence of bursting.

TEST NUMBER 2: BURSTING

If item is not rejected, replace it in fixture, using opposite end of body and apply 192 pounds compression load. If, item shows no evidence of bursting (punch travels thru material) it shall be rejected.

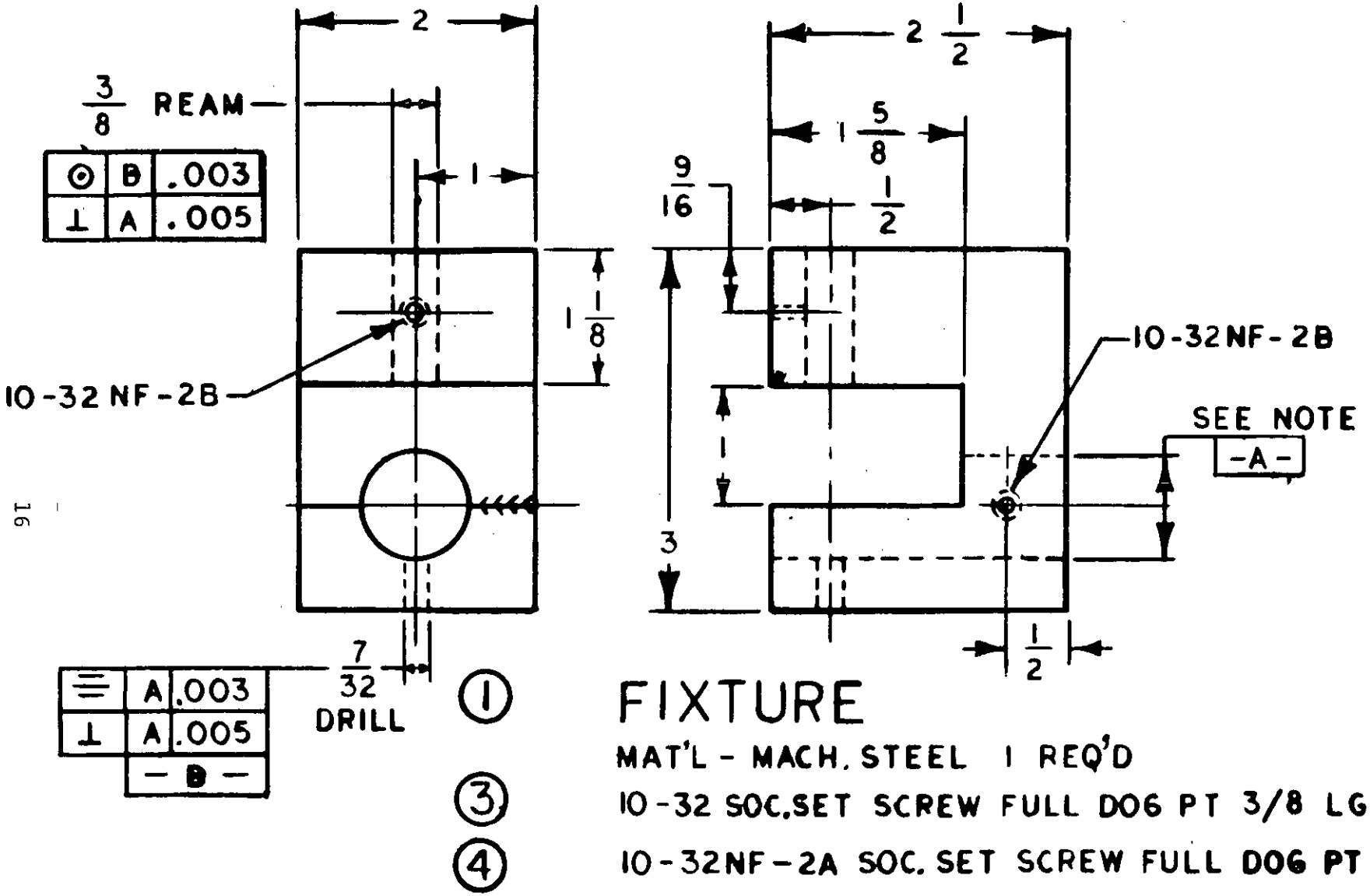
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ASSEMBLY
BUSTING STRENGTH
TEST FIXTURE

FIGURE 1 SHEET 1

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FIXTURE

MAT'L - MACH. STEEL 1 REQ'D

③ 10-32 SOC. SET SCREW FULL DOG PT 3/8 LG

④ 10-32NF-2A SOC. SET SCREW FULL DOG PT 1/2 LG

NOTE DIAMETER TO SUIT APPLICABLE BODY ASS'Y

FIGURE 1 SHEET 2

DIMENSIONS ARE IN INCHES
TOLERANCES ON DICIMALS ±.005
FRACTIONS: 1/64
ANGLE ± 2°

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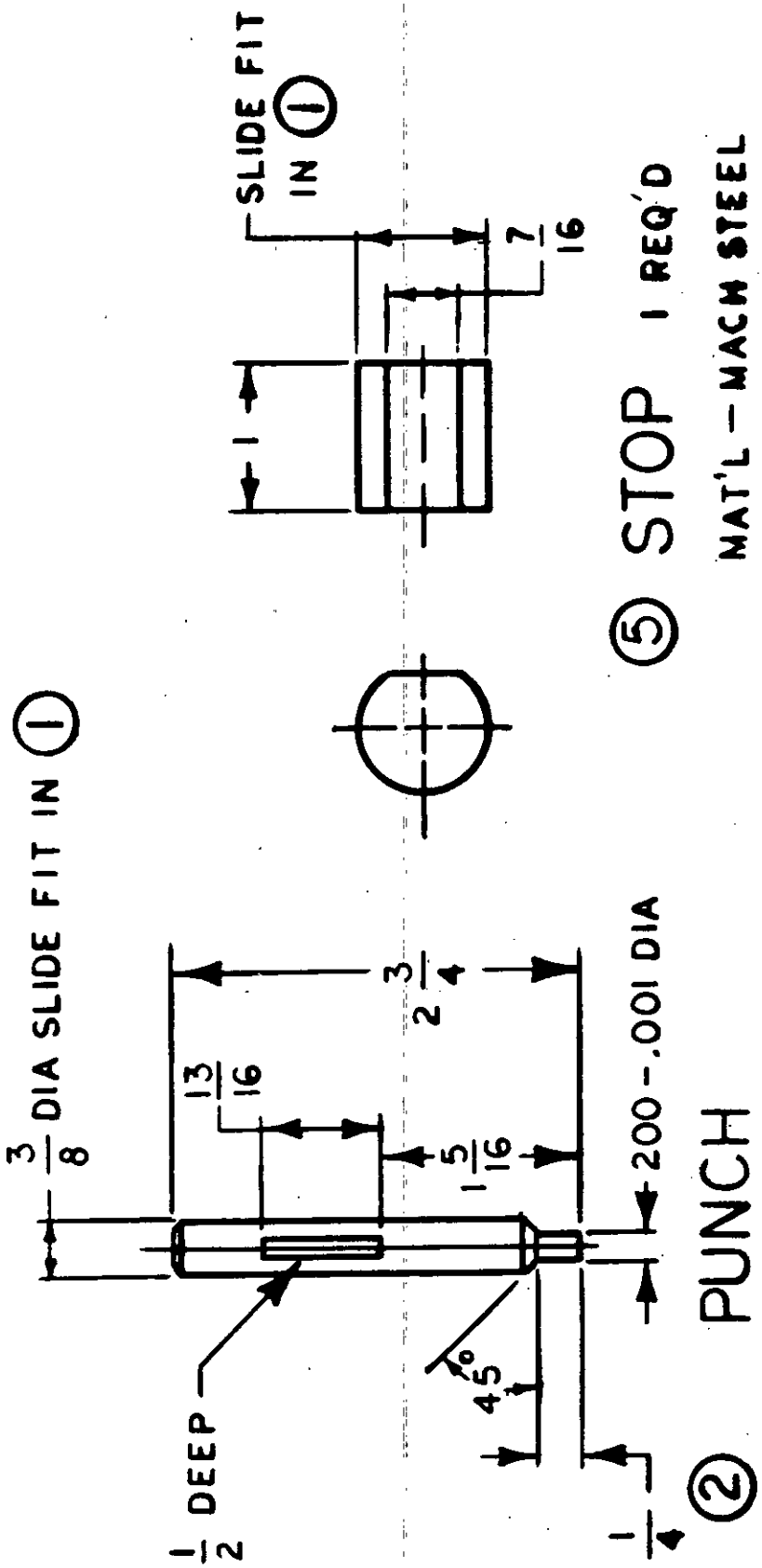


FIGURE 1 SHEET 3

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6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the M702A1 Ignition Cartridge.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: AMSMC-QAT-I, Dover, NJ 07801-5001.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as, ARRADCOM, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-48653	CARTRIDGE, IGNITION, M702A1 PARTS FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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